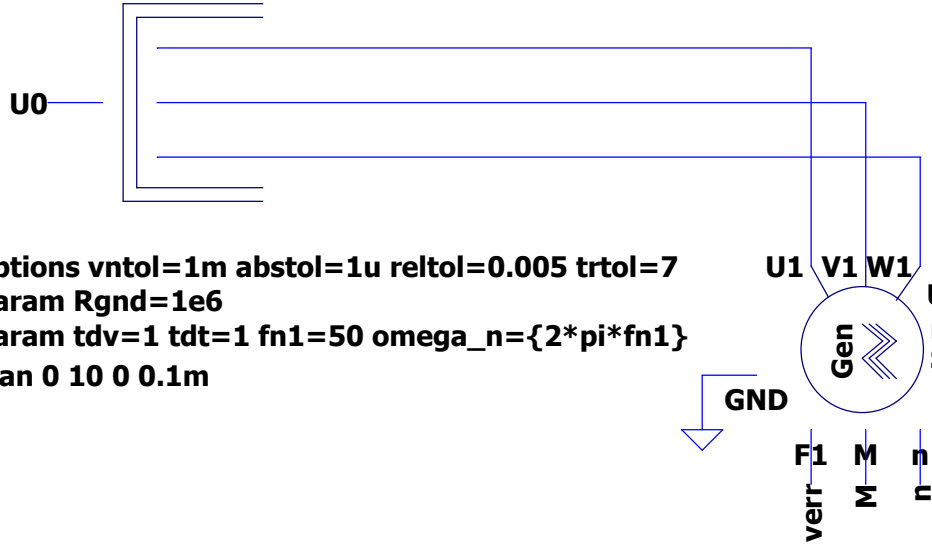
Synchronous 400V generator
Generator has variable magnetic field voltage (0 to 96V)
Generator has fixed speed at 1500
Generator is not loaded
Schenkelpolfaktor=1 (cylindrical pole machine)
a=0.6-0.9 m=7 (nonlinear magnetics - with saturation)



.options vntol=1m abstol=1u reltol=0.005 trtol=7
.param Rgnd=1e6
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.tran 0 10 0 0.1m

UN_verkettet=400V IN=21A Pel=12kW a=0.6 m=7 UE0=48V IE0=5
n0=1500 t_anlauf=2m t_auslauf=2 t_betrieb=100 n_Turbine=1500
Schenkelpolfaktor=1 Daempferfaktor=1 J=1 Streuung=0.05 rcu=0.2 rbg=0.00005

High value to rcu to get
fast steady state

verr verr 0 pwl 0s 0v 10s 96v

Magnetic field voltage
0 to 96V